


# IE3090
# Network Programming
# 3rd Year, Semester I

Submitted to

Sri Lanka Institute of Information Technology

## Design Diary — IE3090 RemoteOps Assignment

In partial fulfillment of the requirements for the

Bachelor of Science Special Honors Degree in Information Technology

5th October 2026

## Declaration

I certify that this report does not incorporate without acknowledgement, any material previously submitted for a degree or diploma in any university, and to the best of my knowledge and belief it does not contain any material previously published or written by another person, except where due reference is made in text.

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**Session 1 — Understanding the brief**

Got all the personal values from IT24101899:

port 9410, SID 9981, token OPS-1899, the log file remoteops_IT24101899.log and the storage location ./agentfiles/IT24101899/. Decided to define each value as a #define at the top of agent_899.c and controller_899.c.

**Session 2 — First Agent version and build**

Wrote the accept loop handling one thread per connection. Had an obstacle due to "missing separator" in the Makefile — spaces not tabs. Verified with cat -A and then fixed in nano."va_list undeclared" — fixed by adding #include .I think this is all the fixes for session 2.

**Session 3 — Protocol commands**

handle_line() is the dispatcher plus AUTH gating. Implemented SYSINFO, LISTPROC, EXEC (using a WHITELIST[] table), PUT and GET. Decided to have a per session rxbuf shared between the line reader and the byte reader, so the leftovers after the PUT header are not lost.This also fixed a bug in the way that a PUT could have the header and the file bytes in the same recv().Also decided that reject_put() should drain the announced bytes before sending an error, so that the framing is not spoiled.

**Session 4 — UDP monitoring and concurrency**

monitor_thread(), start_monitor(), stop_monitor(). Had an obstacle where the 2 second interval for UDP was flooding the Controller — increased MON_INTERVAL_SEC from 2 to 5.Tested this by running 5 Controllers in parallel and they all get the correct replies.

**Session 5 — Stability and polish**

Stability obstacle: SSH drop killed the Agent — reconnected,learned about tmux.Another obstacle: AI suggested `stty -echo` but that messed up my terminal — recovered with `reset`, learned not to do that. logmsg() with timestamp + mutex.Renaming functions so that the code and report use the same names.

**Key decisions**

Thread per connection: simple, robust and meets the 5 client requirement.Single reply() function that adds the SID:9981 in one place.WHITELIST is a constant table — user input never reaches a shell.Validating the filename for PUT to prevent path traversal.

**Obstacles and fixes**

makefile tabs → cat -A + nano,Missing → added include,UDP flooding 2 s → 5 s

SSH disconnect → tmux,stty -echo → reset,PUT byte counting → shared rxbuf

Filename security → validation + reject_put